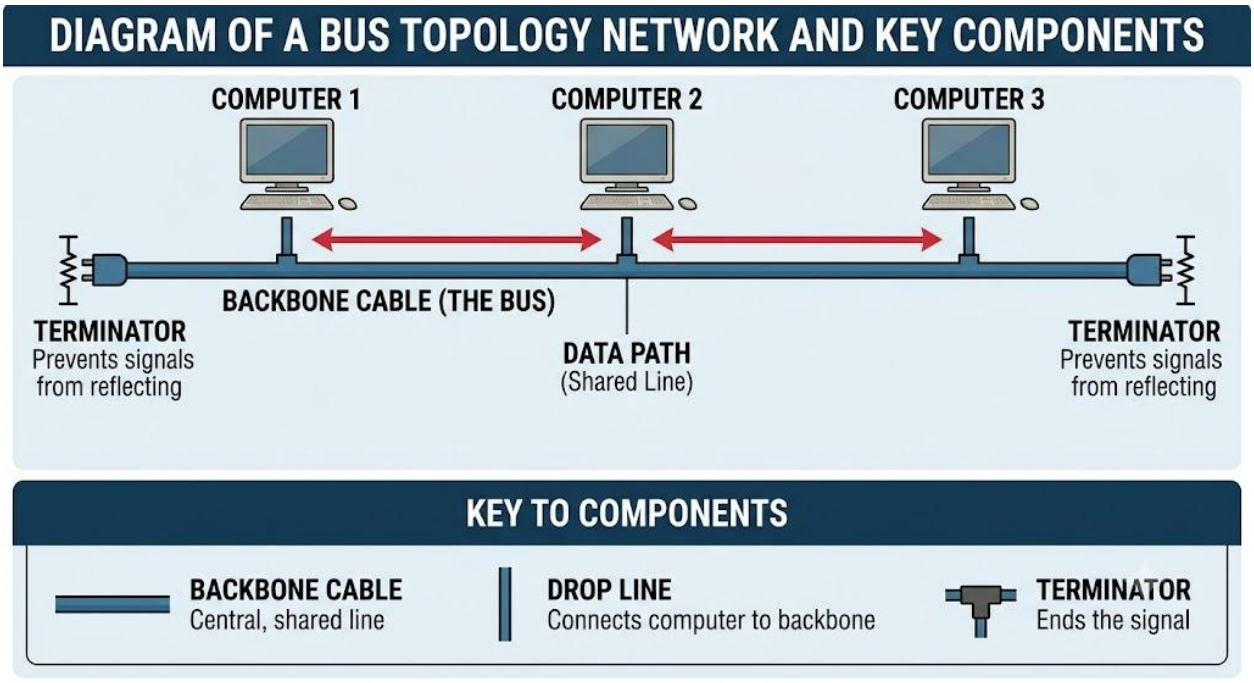
DAY – 2 NETWORK TOPOLOGY

1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

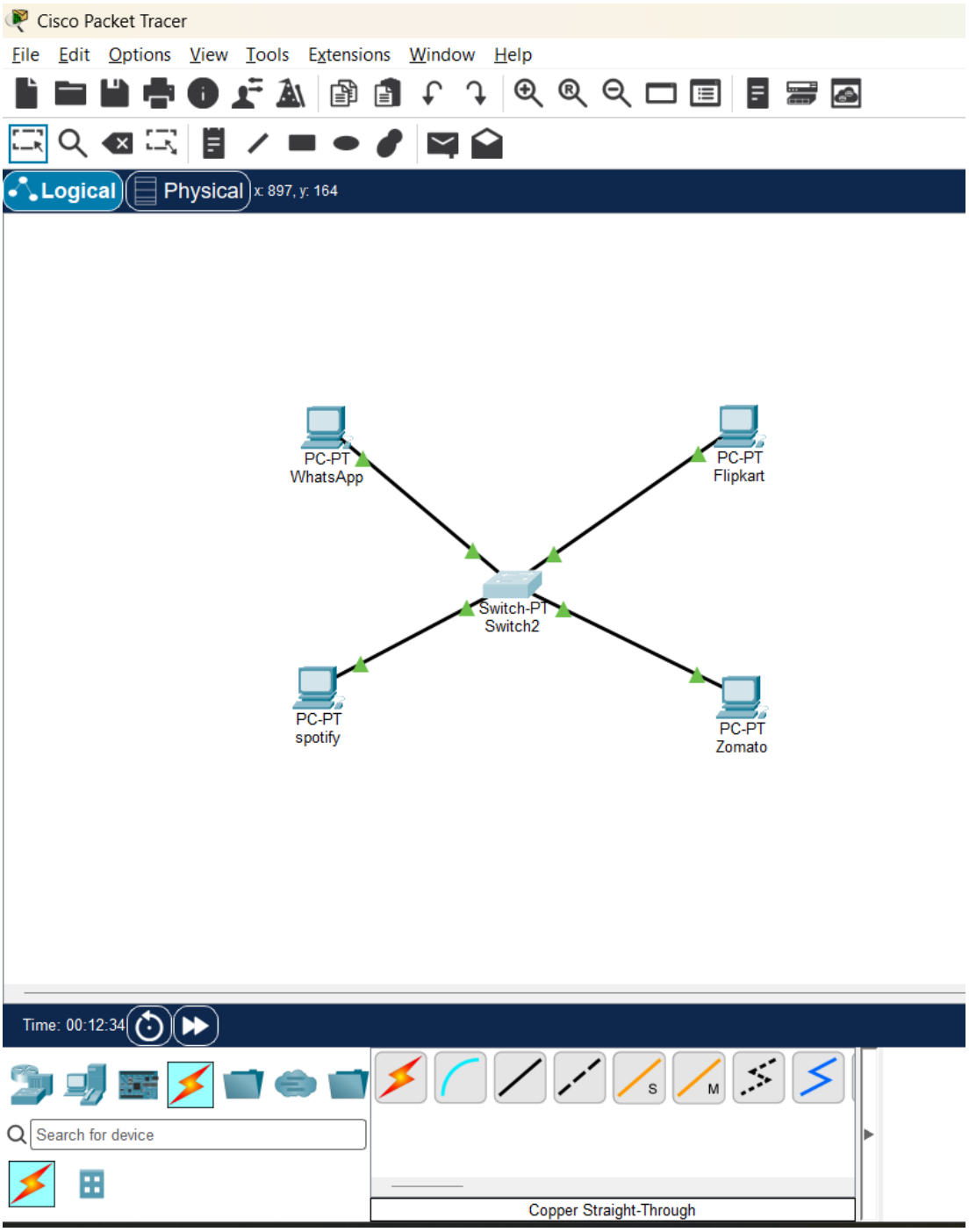
Ans:-



* Real-world example: An old school computer lab where 10 computers are connected using a single coaxial cable. All computers share the same main cable to communicate and exchange data.

2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, and Flipkart) and take a screenshot of your topology.

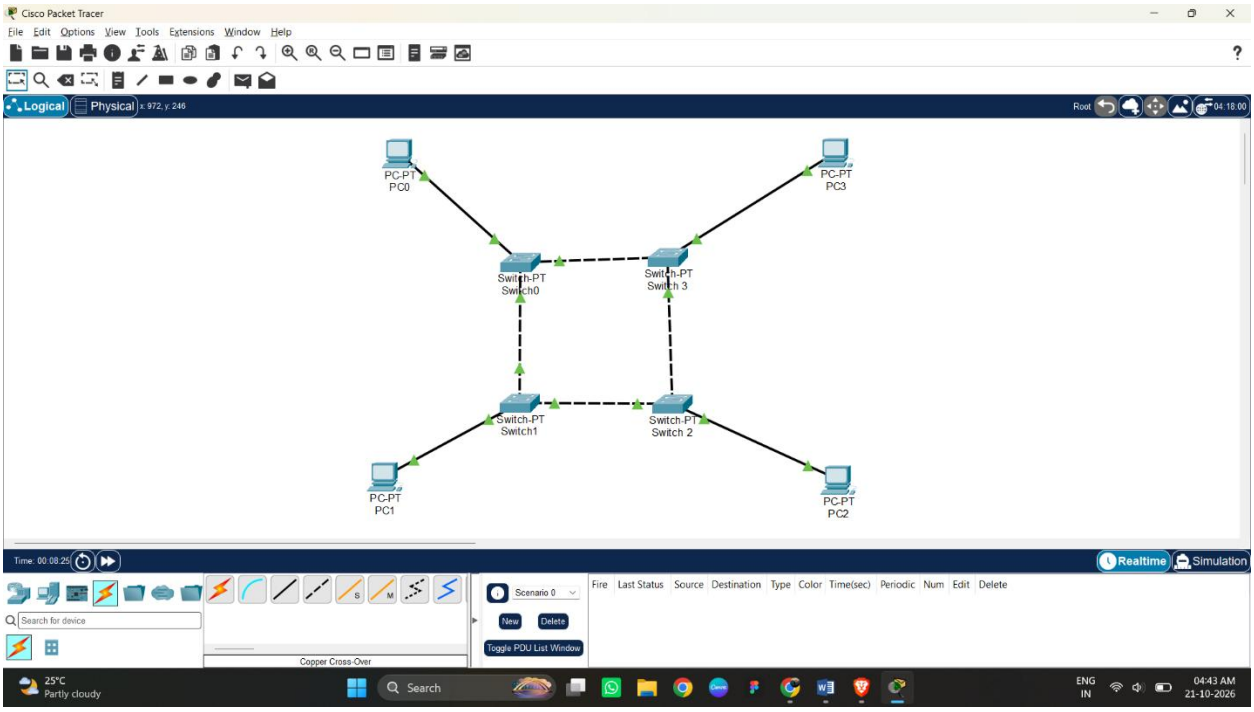
Ans:-



3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Hint: Use arrows to show the direction of data flow.

Ans:-



Explanation: In a ring topology, each switch connects to two neighboring switches, creating a closed loop. Data travels between devices through the network toward its destination.

4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Ans:-

Topology	Advantages	Disadvantages
Star Topology	1. **Easy Troubleshooting & Setup:** Adding new devices or isolating a broken cable is straightforward since each device connects independently to the central switch. 2. **Isolated Failures:** If one cable or device fails, only that specific connection is affected; the rest of the network keeps working.	1. **Single Point of Failure:** If the central hub or switch fails, the entire network goes down. 2. **Cable Dependency:** It requires a significant amount of cable running from every single device back to the central switch.
Mesh Topology	1. **High Reliability & Fault Tolerance:** Because multiple paths exist between devices, data can automatically reroute if a connection fails. 2. **No Single Point of Failure:** The failure of a single cable or node does not disrupt the rest of the network.	1. **High Cost:** Requires a massive amount of cabling and network ports for every device, making it very expensive to build. 2. **Complex Installation:** Setting up and configuring a full mesh network is complex and difficult to manage as it grows.